

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 7

4 Economics, Law and Ethics (ah793)

game theory

- (a) How does game theory account for the distribution of trolls (accounts that make deliberately offensive and provocative posts) on social media platforms?

[10 marks]

Answer: The hawk-dove game models the probability that individuals in a population will be aggressive (*2 marks*). On social media platforms, the hawk-dove game can be applied to understand how trolls seek the limited resources of attention and social status (*2 marks*).

Individuals can choose to adopt a ‘hawk’ strategy, which involves being aggressive and competing for resources, or the ‘dove’ strategy, which involves being co-operative and non-aggressive (*2 marks*). Doves like each others’ posts and provide pro-social comments. Hawks demean doves, and fight other hawks (with a risk of being banned from the platform) (*2 marks*). If the probability of being banned is less than the probability of gaining attention, a small number of hawks will prosper, as most interactions will be with doves (*2 marks*).

information
economics

- (b) If a popular social media platform changes its moderation policy, allowing trolls to proliferate, when are users likely to switch to another platform? [10 marks]

Answer: Users are likely to switch to another platform when the costs of switching become lower than the benefits of staying on the current platform (*2 marks*). Switching costs may include setting up a new account, rebuilding their social network from scratch, and generating levels of ‘esteem’ (e.g. number of ‘followers’ and ‘likes’) (*3 marks*). This will be dependent on the availability of alternative platforms with more favourable moderation policies (*1 mark*). Network externalities mean that these social media platforms become more valuable the more people use them (*2 marks*). Therefore, it is likely switching will only occur if platforms tip, and the majority of users move together, so that users can maintain their existing network of friends, family, and contacts on the alternative platform (*2 marks*).
